

Aim: JDBC Programming.

1. Write a jdbc program to:

i. insert depno, deptname and location of department from java console.

ii. insert depno, deptname and location of department from java console using PreparedStatement.

iii. update department information from java console.

CODE:

```
package jdbc.projects;
import java.sql.*;
import java.io.*;
public class JDBCProjects {
    public static void main(String[] args) {
        try{
            Class.forName("com.mysql.jdbc.Driver");
            try(Connection con = DriverManager.getConnection("jdbc:mysql://localhost/atomic",
"root", "atomic")){
                Statement st = con.createStatement();
                BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
                System.out.print("Enter Department No (Console Insert): ");
                int no = Integer.parseInt(br.readLine());
                System.out.print("Enter Department Name: ");
                String dn = br.readLine();
                System.out.print("Enter the Location: ");
                String loc = br.readLine();
                String sql1 = "INSERT INTO department VALUES(" + no + ", " + dn + ", " + loc + ")";
                st.executeUpdate(sql1);
                System.out.println("Record Saved! (Console Insert)");
                System.out.println("\nEnter DeptNo (PreparedStatement Insert):");
                System.out.print("Enter Department No: ");
                int no1 = Integer.parseInt(br.readLine());
                System.out.print("Enter Department Name: ");
                String dn1 = br.readLine();
                System.out.print("Enter the Location: ");
                String loc1 = br.readLine();
                String sql2 = "INSERT INTO department VALUES (?, ?, ?)";
                PreparedStatement ps = con.prepareStatement(sql2);
                ps.setInt(1, no1);
                ps.setString(2, dn1);
                ps.setString(3, loc1);
                ps.executeUpdate();
                System.out.println("Record Saved! (PreparedStatement Insert)");
                System.out.print("\nEnter Department No for updation (Console Update): ");
                int no2 = Integer.parseInt(br.readLine());
                System.out.print("Enter new Location: ");
                String loc2 = br.readLine();
                String sql3 = "UPDATE department SET location=" + loc2 + " WHERE depno=" + no2;
                st.executeUpdate(sql3);
                System.out.println("Updation Complete!....");
            }
        }
    }
}
```

```

    }
}
catch(ClassNotFoundException | SQLException | IOException | NumberFormatException e){
    System.out.println("Error:" + e.getMessage());
}
}
}

```

OUTPUT:

```

mysql> select * from department;
+-----+-----+-----+
| depno | depname | location |
+-----+-----+-----+
|      1 | CS      | SFC      |
+-----+-----+-----+
1 row in set (0.00 sec)

mysql> select * from department;
+-----+-----+-----+
| depno | depname | location |
+-----+-----+-----+
|      1 | CS      | 2ND FLOOR |
|      2 | IT      | SFC      |
|      3 | BMS     | 4TH FLOOR |
+-----+-----+-----+
3 rows in set (0.00 sec)

```

```

Loading class 'com.mysql.jdbc.Driver'. This is deprecated.
Enter Department No (Console Insert): 2
Enter Department Name: IT
Enter the Location: SFC
Record Saved! (Console Insert)

Enter DeptNo (PreparedStatement Insert):
Enter Department No: 3
Enter Department Name: BMS
Enter the Location: 4TH FLOOR
Record Saved! (PreparedStatement Insert)

Enter Department No for updation (Console Update): 1
Enter new Location: 2ND FLOOR
Updation Complete!...
BUILD SUCCESSFUL (total time: 38 seconds)

```

2. Write a jdbc program to insert name and age of the student into the Student table using the following GUI.

CODE:

```

package jdbc.projects;

import java.sql.*;

public class student_frame extends javax.swing.JFrame {
    private static final java.util.logging.Logger logger = java.util.logging.Logger.getLogger(student_frame.class.getName());
    public student_frame() {
        initComponents();
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {
        jLabel1 = new javax.swing.JLabel();
        jLabel2 = new javax.swing.JLabel();
        jButton1 = new javax.swing.JButton();
        jTextField1 = new javax.swing.JTextField();
        jTextField2 = new javax.swing.JTextField();
        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        jLabel1.setFont(new java.awt.Font("Segoe UI Black", 1, 14)); // NOI18N
        jLabel1.setForeground(new java.awt.Color(102, 102, 255));
        jLabel1.setText("Name:");
        jLabel2.setFont(new java.awt.Font("Segoe UI Black", 1, 14)); // NOI18N
        jLabel2.setForeground(new java.awt.Color(102, 102, 255));
        jLabel2.setText("Age:");
        jButton1.setBackground(new java.awt.Color(204, 204, 255));
        jButton1.setFont(new java.awt.Font("Segoe UI Black", 1, 14)); // NOI18N
        jButton1.setForeground(new java.awt.Color(102, 102, 255));
        jButton1.setText("Submit");
        jButton1.setBorder(javax.swing.BorderFactory.createEtchedBorder());
        jButton1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton1ActionPerformed(evt);
            }
        });
    }
}

```

```

jTextField1.setBackground(new java.awt.Color(204, 204, 255));
jTextField1.setFont(new java.awt.Font("Segoe UI Black", 1, 14)); // NOI18N
jTextField1.setForeground(new java.awt.Color(102, 102, 255));
jTextField1.setBorder(javax.swing.BorderFactory.createEtchedBorder());
jTextField2.setBackground(new java.awt.Color(204, 204, 255));
jTextField2.setFont(new java.awt.Font("Segoe UI Black", 1, 14)); // NOI18N
jTextField2.setForeground(new java.awt.Color(102, 102, 255));
jTextField2.setBorder(javax.swing.BorderFactory.createEtchedBorder());
javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
getContentPane().setLayout(layout);
layout.setHorizontalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addGroup(layout.createSequentialGroup()
                    .addGap(27, 27, 27)
                    .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING, false)
                        .addComponent(jLabel1, javax.swing.GroupLayout.DEFAULT_SIZE, 70, Short.MAX_VALUE)
                        .addComponent(jLabel2, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
                    .addGap(18, 18, 18)
                    .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING, false)
                        .addComponent(jTextField1)
                        .addComponent(jTextField2, javax.swing.GroupLayout.DEFAULT_SIZE, 270, Short.MAX_VALUE)))
                .addGroup(layout.createSequentialGroup()
                    .addGap(218, 218, 218)
                    .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 87, javax.swing.GroupLayout.PREFERRED_SIZE)))
            .addContainerGap(143, Short.MAX_VALUE))
        );
layout.setVerticalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .addGap(96, 96, 96)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                .addComponent(jLabel1, javax.swing.GroupLayout.PREFERRED_SIZE, 32, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE))
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                .addComponent(jLabel2, javax.swing.GroupLayout.PREFERRED_SIZE, 38, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jTextField2, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE))
            .addGap(69, 69, 69)
            .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 38, javax.swing.GroupLayout.PREFERRED_SIZE)
            .addContainerGap(133, Short.MAX_VALUE))
        );

pack();
}

private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
    try{
        Class.forName("com.mysql.jdbc.Driver");
        try(Connection con = DriverManager.getConnection("jdbc:mysql://localhost/atomic",
"root", "atomic")){
            Statement st = con.createStatement();
            String name = jTextField1.getText();
            int age = Integer.parseInt(jTextField2.getText());
            String sql1 = "insert into student values('"+ name + "', '"+ age + "')";
            st.executeUpdate(sql1);
            System.out.println("\n\t\tRecord Saved....");
        }
    }
}

```

```

    }
    catch(ClassNotFoundException | SQLException | NumberFormatException e){
        System.out.println("Error:" + e.getMessage());
    }
}
public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new student_frame().setVisible(true));
}
private javax.swing.JButton jButton1;
private javax.swing.JLabel jLabel1;
private javax.swing.JLabel jLabel2;
private javax.swing.JTextField jTextField1;
private javax.swing.JTextField jTextField2;

```

OUTPUT:

run:

```
Loading class `com.mysql.jdbc.Driver
```

```
Record Saved....
```

name	age
DarkLord	19

1 row in set (0.01 sec)

name	age
DarkLord	19
Atomic	19

2 rows in set (0.00 sec)

3. Write a jdbc program to create a table in the database using the following GUI. (HW)**CODE:**

```

package jdbc.projects;
import java.sql.*;
public class create_table_from_gui extends javax.swing.JFrame {
    private static final java.util.logging.Logger logger = java.util.logging.Logger.getLogger(create_table_from_gui.class.getName());
    public create_table_from_gui() {
        initComponents();
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {
        jButton1 = new javax.swing.JButton();
        jComboBox1 = new javax.swing.JComboBox<>();
        jComboBox2 = new javax.swing.JComboBox<>();
        jComboBox3 = new javax.swing.JComboBox<>();
        jTextField1 = new javax.swing.JTextField();
        jTextField2 = new javax.swing.JTextField();
        jTextField3 = new javax.swing.JTextField();
        jTextField4 = new javax.swing.JTextField();
        jTextField5 = new javax.swing.JTextField();
        jLabel1 = new javax.swing.JLabel();
        jLabel2 = new javax.swing.JLabel();
        jLabel3 = new javax.swing.JLabel();
    }
}

```

```
jLabel4 = new javax.swing.JLabel();
jLabel5 = new javax.swing.JLabel();
jLabel6 = new javax.swing.JLabel();
jLabel7 = new javax.swing.JLabel();
jTextField6 = new javax.swing.JTextField();
jTextField7 = new javax.swing.JTextField();
jLabel8 = new javax.swing.JLabel();
jLabel9 = new javax.swing.JLabel();
jLabel10 = new javax.swing.JLabel();
setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
jButton1.setBackground(new java.awt.Color(204, 204, 255));
jButton1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jButton1.setForeground(new java.awt.Color(102, 102, 255));
jButton1.setText("Create");
jButton1.setBorder(javax.swing.BorderFactory.createEtchedBorder());
jButton1.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton1ActionPerformed(evt);
    }
});
jComboBox1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jComboBox1.setForeground(new java.awt.Color(102, 102, 255));
jComboBox1.setModel(new javax.swing.DefaultComboBoxModel<>(new String[] { "number", "varchar", "char" }));
jComboBox1.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jComboBox1ActionPerformed(evt);
    }
});
jComboBox2.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jComboBox2.setForeground(new java.awt.Color(102, 102, 255));
jComboBox2.setModel(jComboBox1.getModel());
jComboBox3.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jComboBox3.setForeground(new java.awt.Color(102, 102, 255));
jComboBox3.setModel(jComboBox1.getModel());
jTextField1.setBackground(new java.awt.Color(204, 204, 255));
jTextField1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jTextField1.setForeground(new java.awt.Color(102, 102, 255));
jTextField1.setBorder(javax.swing.BorderFactory.createEtchedBorder());
jTextField2.setBackground(new java.awt.Color(204, 204, 255));
jTextField2.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jTextField2.setForeground(new java.awt.Color(102, 102, 255));
jTextField2.setBorder(javax.swing.BorderFactory.createEtchedBorder());
jTextField2.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jTextField2ActionPerformed(evt);
    }
});
jTextField3.setBackground(new java.awt.Color(204, 204, 255));
jTextField3.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jTextField3.setForeground(new java.awt.Color(102, 102, 255));
jTextField3.setBorder(javax.swing.BorderFactory.createEtchedBorder());
jTextField3.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jTextField3ActionPerformed(evt);
    }
});
jTextField4.setBackground(new java.awt.Color(204, 204, 255));
jTextField4.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jTextField4.setForeground(new java.awt.Color(102, 102, 255));
jTextField4.setBorder(javax.swing.BorderFactory.createEtchedBorder());
jTextField5.setBackground(new java.awt.Color(204, 204, 255));
jTextField5.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jTextField5.setForeground(new java.awt.Color(102, 102, 255));
jTextField5.setBorder(javax.swing.BorderFactory.createEtchedBorder());
jLabel1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel1.setForeground(new java.awt.Color(102, 102, 255));
```

```

jLabel1.setText("Column 1:");
jLabel2.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel2.setForeground(new java.awt.Color(102, 102, 255));
jLabel2.setText("Column 2:");
jLabel3.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel3.setForeground(new java.awt.Color(102, 102, 255));
jLabel3.setText("Column 3:");
jLabel4.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel4.setForeground(new java.awt.Color(102, 102, 255));
jLabel4.setText("Data Type:");
jLabel5.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel5.setForeground(new java.awt.Color(102, 102, 255));
jLabel5.setText("Data Type:");
jLabel6.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel6.setForeground(new java.awt.Color(102, 102, 255));
jLabel6.setText("Data Type:");
jLabel7.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel7.setForeground(new java.awt.Color(102, 102, 255));
jLabel7.setText("Size:");
jTextField6.setBackground(new java.awt.Color(204, 204, 255));
jTextField6.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jTextField6.setForeground(new java.awt.Color(102, 102, 255));
jTextField6.setBorder(javax.swing.BorderFactory.createEtchedBorder());
jTextField7.setBackground(new java.awt.Color(204, 204, 255));
jTextField7.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jTextField7.setForeground(new java.awt.Color(102, 102, 255));
jTextField7.setBorder(javax.swing.BorderFactory.createEtchedBorder());
jTextField7.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jTextField7ActionPerformed(evt);
    }
});
jLabel8.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel8.setForeground(new java.awt.Color(102, 102, 255));
jLabel8.setText("Size:");
jLabel9.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel9.setForeground(new java.awt.Color(102, 102, 255));
jLabel9.setText("Size:");
jLabel10.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel10.setForeground(new java.awt.Color(102, 102, 255));
jLabel10.setText("Table Name:");
javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
getContentPane().setLayout(layout);
layout.setHorizontalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(javax.swing.GroupLayout.Alignment.TRAILING, layout.createSequentialGroup()
            .addContainerGap()
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addComponent(jLabel8, javax.swing.GroupLayout.PREFERRED_SIZE, 49, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jLabel7, javax.swing.GroupLayout.PREFERRED_SIZE, 49, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jLabel9, javax.swing.GroupLayout.PREFERRED_SIZE, 49, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addGap(32, 32, 32)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                    .addComponent(jTextField5, javax.swing.GroupLayout.Alignment.TRAILING, javax.swing.GroupLayout.PREFERRED_SIZE, 71, javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(jTextField4, javax.swing.GroupLayout.Alignment.TRAILING, javax.swing.GroupLayout.PREFERRED_SIZE, 71, javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(jTextField7, javax.swing.GroupLayout.Alignment.TRAILING, javax.swing.GroupLayout.PREFERRED_SIZE, 71, javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addGap(84, 84, 84)
                    .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                        .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 86, javax.swing.GroupLayout.PREFERRED_SIZE)
                    )
                )
            )
        )
);

```

```
.addGroup(layout.createSequentialGroup())
.addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
.addGroup(layout.createSequentialGroup())
.addGap(274, 274, 274)
.addComponent(jLabel10)
.addGap(31, 31, 31))
.addGroup(layout.createSequentialGroup())
.addGap(28, 28, 28)
.addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.TRAILING)
.addComponent(jLabel1)
.addComponent(jLabel2)
.addComponent(jLabel3))
.addGap(34, 34, 34)
.addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
.addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
.addComponent(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, 105,
javax.swing.GroupLayout.PREFERRED_SIZE)
.addComponent(jTextField2, javax.swing.GroupLayout.Alignment.TRAILING,
javax.swing.GroupLayout.PREFERRED_SIZE, 105, javax.swing.GroupLayout.PREFERRED_SIZE))
.addComponent(jTextField3, javax.swing.GroupLayout.Alignment.TRAILING,
javax.swing.GroupLayout.PREFERRED_SIZE, 105, javax.swing.GroupLayout.PREFERRED_SIZE))
.addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED, 49, Short.MAX_VALUE)
.addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
.addComponent(jLabel6, javax.swing.GroupLayout.PREFERRED_SIZE, 93,
javax.swing.GroupLayout.PREFERRED_SIZE)
.addComponent(jLabel5, javax.swing.GroupLayout.PREFERRED_SIZE, 81,
javax.swing.GroupLayout.PREFERRED_SIZE)
.addComponent(jLabel4))
.addGap(18, 18, Short.MAX_VALUE)))
.addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
.addComponent(jComboBox2, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
.addComponent(jTextField6, javax.swing.GroupLayout.PREFERRED_SIZE, 124,
javax.swing.GroupLayout.PREFERRED_SIZE)
.addComponent(jComboBox3, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
.addComponent(jComboBox1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)))
.addContainerGap(258, Short.MAX_VALUE))
);
layout.setVerticalGroup(
layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
.addGroup(javax.swing.GroupLayout.Alignment.TRAILING, layout.createSequentialGroup())
.addGap(22, 22, 22)
.addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
.addComponent(jTextField6, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
.addComponent(jLabel10))
.addGap(47, 47, 47)
.addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
.addComponent(jComboBox1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
.addComponent(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
.addComponent(jTextField4, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
.addComponent(jLabel1)
.addComponent(jLabel4)
.addComponent(jLabel7))
.addGap(74, 74, 74)
.addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
.addComponent(jComboBox2, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
.addComponent(jTextField2, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
```

```

        .addComponent(jTextField5, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
        .addComponent(jLabel2)
        .addComponent(jLabel5)
        .addComponent(jLabel8))
        .addGap(70, 70, 70)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
        .addComponent(jComboBox3, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
        .addComponent(jTextField3, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
        .addComponent(jLabel3)
        .addComponent(jLabel6)
        .addComponent(jTextField7, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
        .addComponent(jLabel9))
        .addGap(59, 59, 59)
        .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 34, javax.swing.GroupLayout.PREFERRED_SIZE)
        .addContainerGap(127, Short.MAX_VALUE)
    );
    pack();
}
private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
    try{
        Class.forName("com.mysql.jdbc.Driver");
        try(Connection con = DriverManager.getConnection("jdbc:mysql://localhost/atomic",
"root", "atomic")){
            Statement st = con.createStatement();
            String table_name = jTextField6.getText();
            String c1 = jTextField1.getText();
            String c2 = jTextField2.getText();
            String c3 = jTextField3.getText();
            int s1 = Integer.parseInt(jTextField4.getText());
            int s2 = Integer.parseInt(jTextField5.getText());
            int s3 = Integer.parseInt(jTextField7.getText());
            String data_type1 = (String) jComboBox1.getSelectedItem();
            String data_type2 = (String) jComboBox2.getSelectedItem();
            String data_type3 = (String) jComboBox3.getSelectedItem();
            String sql1 = "CREATE TABLE `" + table_name + "` (" +
                    "`" + c1 + "` " + data_type1 + "(" + s1 + ")," +
                    "`" + c2 + "` " + data_type2 + "(" + s2 + ")," +
                    "`" + c3 + "` " + data_type3 + "(" + s3 + ")" +
                    ");"
            st.executeUpdate(sql1);
            System.out.println("Table created....");
        }
    }
    catch(ClassNotFoundException | SQLException | NumberFormatException e){
        System.out.println("Error:" + e.getMessage());
    }
}
public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new create_table_from_gui().setVisible(true));
}
private javax.swing.JButton jButton1;
private javax.swing.JComboBox<String> jComboBox1;

```



```

private javax.swing.JComboBox<String> jComboBox2;
private javax.swing.JComboBox<String> jComboBox3;
private javax.swing.JLabel jLabel1;
private javax.swing.JLabel jLabel10;
private javax.swing.JLabel jLabel2;
private javax.swing.JLabel jLabel3;
private javax.swing.JLabel jLabel4;
private javax.swing.JLabel jLabel5;
private javax.swing.JLabel jLabel6;
private javax.swing.JLabel jLabel7;
private javax.swing.JLabel jLabel8;
private javax.swing.JLabel jLabel9;
private javax.swing.JTextField jTextField1;
private javax.swing.JTextField jTextField2;
private javax.swing.JTextField jTextField3;
private javax.swing.JTextField jTextField4;
private javax.swing.JTextField jTextField5;
private javax.swing.JTextField jTextField6;
private javax.swing.JTextField jTextField7;
}

```

OUTPUT:

Table Name:

Column 1: DataType: Size:

Column 2: DataType: Size:

Column 3: DataType: Size:

run:

```

Loading class `com.mysql.jdbc.Driver'.
Table created....

```

```

mysql> desc student_one;
+-----+-----+-----+-----+-----+-----+
| Field | Type   | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| sname | char(30) | YES  |     | NULL    |       |
| sage  | int      | YES  |     | NULL    |       |
| snumber | int      | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.01 sec)

```

4. Write a JDBC program to delete an employee record from the Employee table using the following GUI. (HW)

CODE:

```

package jdbc.projects;
import java.sql.*;
import java.sql.DriverManager;
import java.sql.SQLException;
import java.sql.Statement;
public class gui_employee_delete extends javax.swing.JFrame {
    private static final java.util.logging.Logger logger = java.util.logging.Logger.getLogger(gui_employee_delete.class.getName());
    public gui_employee_delete() {
        initComponents();
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {

```

```

jButton1 = new javax.swing.JButton();
jLabel1 = new javax.swing.JLabel();
jTextField1 = new javax.swing.JTextField();
setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
jButton1.setBackground(new java.awt.Color(204, 204, 255));
jButton1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jButton1.setForeground(new java.awt.Color(102, 102, 255));
jButton1.setText("Delete");
jButton1.setBorder(javax.swing.BorderFactory.createEtchedBorder());
jButton1.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton1ActionPerformed(evt);
    }
});
jLabel1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jLabel1.setForeground(new java.awt.Color(102, 102, 255));
jLabel1.setText("Enter Employee ID:");
jTextField1.setBackground(new java.awt.Color(204, 204, 255));
jTextField1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
jTextField1.setForeground(new java.awt.Color(102, 102, 255));
jTextField1.setBorder(javax.swing.BorderFactory.createEtchedBorder());
javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
getContentPane().setLayout(layout);
layout.setHorizontalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addGroup(layout.createSequentialGroup()
                    .add(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                        .add(jLabel1)
                        .add(jTextField1)
                        .add(jButton1))
                    .addGap(10, 10, 10))
                .addGroup(layout.createSequentialGroup()
                    .add(jLabel1)
                    .addGap(10, 10, 10))
                .addGroup(layout.createSequentialGroup()
                    .add(jButton1)
                    .addGap(10, 10, 10))
            )
        )
);
layout.setVerticalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .add(jLabel1)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .add(jTextField1)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .add(jButton1)
        )
);
pack();
}

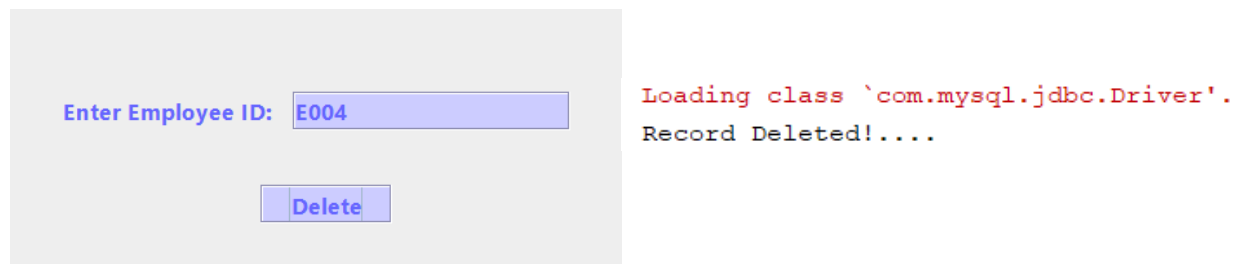
private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
    try {
        Class.forName("com.mysql.jdbc.Driver");
        try (Connection con = DriverManager.getConnection("jdbc:mysql://localhost/atomic",
            "root", "atomic")) {
            Statement st = con.createStatement();
            String eid = jTextField1.getText();
            String sql1 = "delete from employee where eno='" + eid + "'";
            st.executeUpdate(sql1);
        }
    } catch (Exception e) {
        e.printStackTrace();
    }
}

```

```

        System.out.println("Record Deleted!....");
    }
}
catch(ClassNotFoundException | SQLException | NumberFormatException e){
    System.out.println("Error:" + e.getMessage());
}
}
}
public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new gui_employee_delete().setVisible(true));
}
private javax.swing.JButton jButton1;
private javax.swing.JLabel jLabel1;
private javax.swing.JTextField jTextField1;
}

```

OUTPUT:


Enter Employee ID:

Loading class `com.mysql.jdbc.Driver`.
Record Deleted!....

```

mysql> select * from employee;
+-----+-----+-----+
| eno   | salary | department |
+-----+-----+-----+
| E001  | 35000  | Sales      |
| E002  | 45000  | HR         |
| E003  | 55000  | IT         |
| E005  | 40000  | Admin      |
+-----+-----+-----+
4 rows in set (0.00 sec)

```

5. Write a JDBC program to select the maximum salary of the employee from the employee table using the following GUI.

CODE:

```

package jdbc.projects;

import java.sql.*;

public class gui_employee_max extends javax.swing.JFrame {
    private static final java.util.logging.Logger logger = java.util.logging.Logger.getLogger(gui_employee_max.class.getName());
    public gui_employee_max() {
        initComponents();
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {
        jButton1 = new javax.swing.JButton();
        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        jButton1.setBackground(new java.awt.Color(204, 204, 255));
        jButton1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
        jButton1.setForeground(new java.awt.Color(102, 102, 255));
        jButton1.setText("Find Max Salary");
        jButton1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton1ActionPerformed(evt);
            }
        });
    }
}

```

```

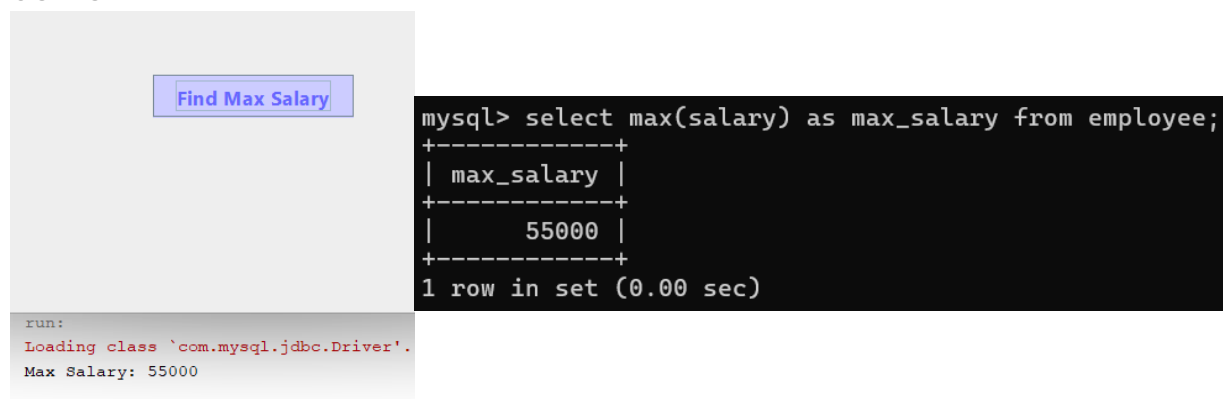
javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
getContentPane().setLayout(layout);
layout.setHorizontalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .addGap(125, 125, 125)
            .addComponent(jButton1)
            .addGap(137, Short.MAX_VALUE))
        );
layout.setVerticalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .addGap(135, 135, 135)
            .addComponent(jButton1)
            .addGap(137, Short.MAX_VALUE))
        );
pack();
}

private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
    try{
        Class.forName("com.mysql.jdbc.Driver");
        try(Connection con = DriverManager.getConnection("jdbc:mysql://localhost/atomic",
"root", "atomic")){
            Statement st = con.createStatement();
            int max;
            String sql1 = "select max(salary) as max_salary from employee;";
            ResultSet rs = st.executeQuery(sql1);
            if (rs.next()) {
                max = rs.getInt("max_salary");
                System.out.println("Max Salary: " + max);
            }
        }
    } catch(ClassNotFoundException | SQLException | NumberFormatException e){
        System.out.println("Error:" + e.getMessage());
    }
}

public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new gui_employee_max().setVisible(true));
}

private javax.swing.JButton jButton1;
}

```

OUTPUT:


The screenshot displays a Java Swing window with a single button labeled "Find Max Salary". Below the window, a terminal window shows the execution of a SQL query: `mysql> select max(salary) as max_salary from employee;`. The query result is shown as a table with one row: `max_salary` with a value of `55000`. The terminal also shows the output of the Java application: `run: Loading class 'com.mysql.jdbc.Driver'. Max Salary: 55000`.

6. Write a JDBC program to update the password in the Login table using the following GUI.


```

        .addGap(37, 37, 37))
        .addGroup(javax.swing.GroupLayout.Alignment.TRAILING, layout.createSequentialGroup()
            .addComponent(jLabel3)
            .addGap(25, 25, 25)))
        .addGap(54, 54, 54)
        .addLayout(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.TRAILING, false)
            .addComponent(jTextField1, javax.swing.GroupLayout.DEFAULT_SIZE, 144, Short.MAX_VALUE)
            .addComponent(jTextField2)
            .addComponent(jTextField3)))
        .addGroup(layout.createSequentialGroup()
            .addGap(180, 180, 180)
            .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 75,
javax.swing.GroupLayout.PREFERRED_SIZE)))
        .addContainerGap(70, Short.MAX_VALUE))
    );
    layout.setVerticalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addGap(79, 79, 79)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                    .addComponent(jLabel1)
                    .addComponent(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
                .addGap(43, 43, 43)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                    .addComponent(jLabel2)
                    .addComponent(jTextField2, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
                .addGap(42, 42, 42)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                    .addComponent(jLabel3)
                    .addComponent(jTextField3, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
                .addGap(40, 40, 40)
                .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 31, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addContainerGap(56, Short.MAX_VALUE))
            );
    pack();
}
private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
    try{
        Class.forName("com.mysql.jdbc.Driver");
        try(Connection con = DriverManager.getConnection("jdbc:mysql://localhost/atomic",
"root", "atomic")){
            Statement st = con.createStatement();
            String eid = jTextField1.getText();
            String pd = jTextField2.getText();
            String npd = jTextField3.getText();
            String sql1 = "update employee set password='"+ npd +"' where eno='"+ eid +"'";
            st.executeUpdate(sql1);
            System.out.println("Query Updated!!!");
        }
    }
    catch(ClassNotFoundException | SQLException | NumberFormatException e){
        System.out.println("Error: " + e.getMessage());
    }
}
public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new gui_update_employee_password().setVisible(true));
}

```

```

private javax.swing.JButton jButton1;
private javax.swing.JLabel jLabel1;
private javax.swing.JLabel jLabel2;
private javax.swing.JLabel jLabel3;
private javax.swing.JTextField jTextField1;
private javax.swing.JTextField jTextField2;
private javax.swing.JTextField jTextField3;
}

```

OUTPUT:

Employee ID:

Password:

New Password:

```
Loading class `com.mysql.jdbc.Driver'.
Query Updated!!
```

```
mysql> select * from employee;
+-----+-----+-----+-----+
| eno | salary | department | password |
+-----+-----+-----+-----+
| E001 | 35000 | Sales      | 123      |
| E002 | 45000 | HR         | 456      |
| E003 | 55000 | IT         | 789      |
| E004 | 60000 | Finance    | 101      |
| E005 | 40000 | Admin      | 540      |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> select * from employee;
+-----+-----+-----+-----+
| eno | salary | department | password |
+-----+-----+-----+-----+
| E001 | 35000 | Sales      | 123      |
| E002 | 45000 | HR         | 456      |
| E003 | 55000 | IT         | 7878     |
| E004 | 60000 | Finance    | 101      |
| E005 | 40000 | Admin      | 540      |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

7. Write a JDBC program to insert the student information into the tabular form.

CODE:

```

package jdbc.projects;
import java.sql.*;
import javax.swing.table.DefaultTableModel;

public class insert_student extends javax.swing.JFrame {
    private static final java.util.logging.Logger logger = java.util.logging.Logger.getLogger(insert_student.class.getName());
    public insert_student() {
        initComponents();
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {
        jScrollPane1 = new javax.swing.JScrollPane();
        jTable1 = new javax.swing.JTable();
        jButton1 = new javax.swing.JButton();
        jButton2 = new javax.swing.JButton();
        jLabel1 = new javax.swing.JLabel();
        jLabel2 = new javax.swing.JLabel();
        jLabel3 = new javax.swing.JLabel();
        jLabel4 = new javax.swing.JLabel();
        jTextField1 = new javax.swing.JTextField();
        jTextField2 = new javax.swing.JTextField();
        jTextField3 = new javax.swing.JTextField();
        jTextField4 = new javax.swing.JTextField();
    }
}

```

```
setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
jTable1.setBackground(new java.awt.Color(204, 204, 255));
jTable1.setBorder(javax.swing.BorderFactory.createEtchedBorder());
jTable1.setFont(new java.awt.Font("Segoe UI Black", 1, 12)); // NOI18N
jTable1.setForeground(new java.awt.Color(102, 102, 255));
jTable1.setModel(new javax.swing.table.DefaultTableModel(
    new Object [][] {
        {null, null, null, null},
        {null, null, null, null},
        {null, null, null, null},
        {null, null, null, null}
    },
    new String [] {
        "Title 1", "Title 2", "Title 3", "Title 4"
    }
));
jScrollPane1.setViewportView(jTable1);
jButton1.setBackground(new java.awt.Color(204, 204, 255));
jButton1.setFont(new java.awt.Font("Segoe UI Black", 1, 14)); // NOI18N
jButton1.setForeground(new java.awt.Color(102, 102, 255));
jButton1.setText("Insert");
jButton1.setBorder(javax.swing.BorderFactory.createEtchedBorder());
jButton1.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton1ActionPerformed(evt);
    }
});
jButton2.setBackground(new java.awt.Color(204, 204, 255));
jButton2.setFont(new java.awt.Font("Segoe UI Black", 1, 14)); // NOI18N
jButton2.setForeground(new java.awt.Color(102, 102, 255));
jButton2.setText("Display");
jButton2.setBorder(javax.swing.BorderFactory.createEtchedBorder());
jButton2.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton2ActionPerformed(evt);
    }
});
jLabel1.setFont(new java.awt.Font("Segoe UI Black", 1, 12)); // NOI18N
jLabel1.setForeground(new java.awt.Color(102, 102, 255));
jLabel1.setText("Name:");
jLabel2.setFont(new java.awt.Font("Segoe UI Black", 1, 12)); // NOI18N
jLabel2.setForeground(new java.awt.Color(102, 102, 255));
jLabel2.setText("Age:");
jLabel3.setFont(new java.awt.Font("Segoe UI Black", 1, 12)); // NOI18N
jLabel3.setForeground(new java.awt.Color(102, 102, 255));
jLabel3.setText("Marks:");
jLabel4.setFont(new java.awt.Font("Segoe UI Black", 1, 12)); // NOI18N
jLabel4.setForeground(new java.awt.Color(102, 102, 255));
jLabel4.setText("Grade:");
jTextField1.setBackground(new java.awt.Color(204, 204, 255));
jTextField1.setFont(new java.awt.Font("Segoe UI Black", 1, 12)); // NOI18N
jTextField1.setForeground(new java.awt.Color(102, 102, 255));
jTextField1.setBorder(javax.swing.BorderFactory.createEtchedBorder());
jTextField2.setBackground(new java.awt.Color(204, 204, 255));
jTextField2.setFont(new java.awt.Font("Segoe UI Black", 1, 12)); // NOI18N
jTextField2.setForeground(new java.awt.Color(102, 102, 255));
jTextField2.setBorder(javax.swing.BorderFactory.createEtchedBorder());
jTextField2.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jTextField2ActionPerformed(evt);
    }
});
jTextField3.setBackground(new java.awt.Color(204, 204, 255));
jTextField3.setFont(new java.awt.Font("Segoe UI Black", 1, 12)); // NOI18N
jTextField3.setForeground(new java.awt.Color(102, 102, 255));
```



```

jTextField3.setBorder(javax.swing.BorderFactory.createEtchedBorder());
jTextField4.setBackground(new java.awt.Color(204, 204, 255));
jTextField4.setFont(new java.awt.Font("Segoe UI Black", 1, 12)); // NOI18N
jTextField4.setForeground(new java.awt.Color(102, 102, 255));
jTextField4.setBorder(javax.swing.BorderFactory.createEtchedBorder());
javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
getContentPane().setLayout(layout);
layout.setHorizontalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .addGap(32, 32, 32)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING, false)
                .addComponent(jLabel1, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addComponent(jLabel2, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addComponent(jLabel3, javax.swing.GroupLayout.DEFAULT_SIZE, 48, Short.MAX_VALUE)
                .addComponent(jLabel4, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
            .addGap(18, 18, 18)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addGroup(layout.createSequentialGroup()
                    .addComponent(jTextField1, javax.swing.GroupLayout.DEFAULT_SIZE, 143, Short.MAX_VALUE)
                    .addComponent(jTextField2)
                    .addComponent(jTextField3)
                    .addComponent(jTextField4)
                    .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED, 51, Short.MAX_VALUE)
                    .addComponent(jScrollPane1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addGap(21, 21, 21)
                    .addGroup(layout.createSequentialGroup()
                        .addGap(42, 42, 42)
                        .addComponent(jButton2, javax.swing.GroupLayout.PREFERRED_SIZE, 73, javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))))
            .addContainerGap())
        .addGroup(layout.createSequentialGroup()
            .addGap(19, 19, 19)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addComponent(jScrollPane1, javax.swing.GroupLayout.PREFERRED_SIZE, 184, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addGroup(layout.createSequentialGroup()
                    .addComponent(jLabel1)
                    .addComponent(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addGap(20, 20, 20)
                    .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                        .addComponent(jLabel2)
                        .addComponent(jTextField2, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addGap(22, 22, 22)
                        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                            .addComponent(jLabel3)
                            .addComponent(jTextField3, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                            .addGap(21, 21, 21)
                            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                                .addComponent(jLabel4)
                                .addComponent(jTextField4, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                                .addGap(135, 135, 135)
                                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                                    .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 44, javax.swing.GroupLayout.PREFERRED_SIZE)
                                    .addComponent(jButton2, javax.swing.GroupLayout.PREFERRED_SIZE, 44, javax.swing.GroupLayout.PREFERRED_SIZE))))))
            .addContainerGap())
    );
layout.setVerticalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .addGap(19, 19, 19)
            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                .addComponent(jScrollPane1, javax.swing.GroupLayout.PREFERRED_SIZE, 184, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addGroup(layout.createSequentialGroup()
                    .addComponent(jLabel1)
                    .addComponent(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addGap(20, 20, 20)
                    .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                        .addComponent(jLabel2)
                        .addComponent(jTextField2, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addGap(22, 22, 22)
                        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                            .addComponent(jLabel3)
                            .addComponent(jTextField3, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                            .addGap(21, 21, 21)
                            .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                                .addComponent(jLabel4)
                                .addComponent(jTextField4, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                                .addGap(135, 135, 135)
                                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                                    .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 44, javax.swing.GroupLayout.PREFERRED_SIZE)
                                    .addComponent(jButton2, javax.swing.GroupLayout.PREFERRED_SIZE, 44, javax.swing.GroupLayout.PREFERRED_SIZE))))))
            .addContainerGap())
    );

```

```
.addContainerGap(170, Short.MAX_VALUE))
);
pack();
}
private void jButton2ActionPerformed(java.awt.event.ActionEvent evt) {
    try{
        Class.forName("com.mysql.jdbc.Driver");
        try(Connection con = DriverManager.getConnection("jdbc:mysql://localhost/atomic",
"root", "atomic")){
            Statement st = con.createStatement();
            String head[] = {"Name", "Age", "Marks", "Grade"};
            String data[][] = new String[20][4];
            String str = "select * from student_info";
            ResultSet rs = st.executeQuery(str);
            int i=0;
            while(rs.next()){
                data[i][0] = rs.getString(1);
                data[i][1] = rs.getString(2);
                data[i][2] = rs.getString(3);
                data[i][3] = rs.getString(4);
                i++;
            }
            jTable1.setModel(new DefaultTableModel(data,head));
            jTable1.setVisible(true);
        }
    }
    catch(ClassNotFoundException | SQLException | NumberFormatException e){
        System.out.println("Error:" + e.getMessage());
    }
}
private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
    try{
        Class.forName("com.mysql.jdbc.Driver");
        try(Connection con = DriverManager.getConnection("jdbc:mysql://localhost/atomic",
"root", "atomic")){
            Statement st = con.createStatement();
            String name = jTextField1.getText();
            int age = Integer.parseInt(jTextField2.getText());
            int marks = Integer.parseInt(jTextField3.getText());
            String grade = jTextField4.getText();
            String sql1 = "insert into student_info values('"+ name + "', '"+ age + "', '"+ marks + "',
"'+ grade + "')";
            st.executeUpdate(sql1);
            System.out.println("\n\t\tRecord Saved....");

        }
    }
    catch(ClassNotFoundException | SQLException | NumberFormatException e){
        System.out.println("Error:" + e.getMessage());
    }
}
```

```

    }
}
public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new insert_student().setVisible(true));
}
// Variables declaration - do not modify
private javax.swing.JButton jButton1;
private javax.swing.JButton jButton2;
private javax.swing.JLabel jLabel1;
private javax.swing.JLabel jLabel2;
private javax.swing.JLabel jLabel3;
private javax.swing.JLabel jLabel4;
private javax.swing.JScrollPane jScrollPane1;
private javax.swing.JTable jTable1;
private javax.swing.JTextField jTextField1;
private javax.swing.JTextField jTextField2;
private javax.swing.JTextField jTextField3;
private javax.swing.JTextField jTextField4;
// End of variables declaration
}

```

OUTPUT:

run:

Loading class 'com.mysql.jdbc.Driver'. This is deprecated. The ne

Record Saved....

Name	Age	Marks	Grade
Atomic	18	90	S
DarkLord	19	80	S
Shadow46	18	85	S
Venom	18	85	S
Ghost	19	100	S
Hashmukh	19	86	A
Cloudy_Boss	19	99	S

```

mysql> select * from student_info;
+-----+-----+-----+-----+
| name   | age  | marks | grade |
+-----+-----+-----+-----+
| Atomic | 18   | 90    | S     |
| DarkLord | 19  | 80    | S     |
| Shadow46 | 18  | 85    | S     |
| Venom  | 18   | 85    | S     |
| Ghost  | 19   | 100   | S     |
| Hashmukh | 19  | 86    | A     |
+-----+-----+-----+-----+
6 rows in set (0.10 sec)

mysql> select * from student_info;
+-----+-----+-----+-----+
| name   | age  | marks | grade |
+-----+-----+-----+-----+
| Atomic | 18   | 90    | S     |
| DarkLord | 19  | 80    | S     |
| Shadow46 | 18  | 85    | S     |
| Venom  | 18   | 85    | S     |
| Ghost  | 19   | 100   | S     |
| Hashmukh | 19  | 86    | A     |
| Cloudy_Boss | 19 | 99    | S     |
+-----+-----+-----+-----+
7 rows in set (0.00 sec)

```

8. Write a JDBC program to insert name and age into Student table with prepared statement using following GUI. (HW)

CODE:

```
package jdbc.projects;
```

import java.sql.*;

```

public class prepared_student_insert extends javax.swing.JFrame {
    private static final java.util.logging.Logger logger = java.util.logging.Logger.getLogger(prepared_student_insert.class.getName());
    public prepared_student_insert() {
        initComponents();
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {
        jLabel1 = new javax.swing.JLabel();
        jLabel2 = new javax.swing.JLabel();
        jTextField1 = new javax.swing.JTextField();
        jTextField2 = new javax.swing.JTextField();
        jButton1 = new javax.swing.JButton();
        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        jLabel1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
        jLabel1.setForeground(new java.awt.Color(102, 102, 255));
        jLabel1.setText("Name:");
        jLabel2.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
        jLabel2.setForeground(new java.awt.Color(102, 102, 255));
        jLabel2.setText("Age:");
        jTextField1.setBackground(new java.awt.Color(204, 204, 255));
        jTextField1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
        jTextField1.setForeground(new java.awt.Color(102, 102, 255));
        jTextField1.setBorder(javax.swing.BorderFactory.createEtchedBorder());
        jTextField2.setBackground(new java.awt.Color(204, 204, 255));
        jTextField2.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
        jTextField2.setForeground(new java.awt.Color(102, 102, 255));
        jTextField2.setBorder(javax.swing.BorderFactory.createEtchedBorder());
        jButton1.setBackground(new java.awt.Color(204, 204, 255));
        jButton1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
        jButton1.setForeground(new java.awt.Color(102, 102, 255));
        jButton1.setText("Submit");
        jButton1.setBorder(javax.swing.BorderFactory.createEtchedBorder());
        jButton1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton1ActionPerformed(evt);
            }
        });
        javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addGroup(layout.createSequentialGroup()
                    .add(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                        .add(jLabel1)
                        .add(jLabel2)
                    )
                    .add(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                        .add(jTextField1)
                        .add(jTextField2)
                    )
                    .add(jButton1)
                )
        );
        layout.setVerticalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addGroup(layout.createSequentialGroup()
                    .add(jLabel1)
                    .add(jTextField1)
                    .add(jButton1)
                    .add(jLabel2)
                    .add(jTextField2)
                )
        );
    }
}

```

```

        .addGap(56, 56, 56)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING, false)
            .addComponent(jLabel2, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
            .addComponent(jTextField2, javax.swing.GroupLayout.DEFAULT_SIZE, 37, Short.MAX_VALUE))
        .addGap(53, 53, 53)
        .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 32, javax.swing.GroupLayout.PREFERRED_SIZE)
        .addContainerGap(59, Short.MAX_VALUE)
    );
    pack();
}
private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
    try{
        Class.forName("com.mysql.jdbc.Driver");
        try(Connection con = DriverManager.getConnection("jdbc:mysql://localhost/atomic",
"root", "atomic")){
            Statement st = con.createStatement();
            String name = jTextField1.getText();
            int age = Integer.parseInt(jTextField2.getText());
            String sql1 = "insert into prepared_student values(?,?)";
            PreparedStatement ps = con.prepareStatement(sql1);
            ps.setString(1,name);
            ps.setInt(2, age);
            ps.executeUpdate();
            System.out.println("Inserted!!!");
            ps.close();
            con.close();
        }
    }
    catch(ClassNotFoundException | SQLException | NumberFormatException e){
        System.out.println("Error:" + e.getMessage());
    }
}
public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new prepared_student_insert().setVisible(true));
}
private javax.swing.JButton jButton1;
private javax.swing.JLabel jLabel1;
private javax.swing.JLabel jLabel2;
private javax.swing.JTextField jTextField1;
private javax.swing.JTextField jTextField2;
}

```

OUTPUT:

Name:

Age:

JDBC Projects (run) x JDBC Projects (run) #2

run:
Loading class `com.mysql.jdbc.Driver'.
Inserted!!

```
mysql> select * from prepared_student;
+-----+-----+
| name  | age  |
+-----+-----+
| Atomic | 19   |
+-----+-----+
1 row in set (0.00 sec)

mysql> select * from prepared_student;
+-----+-----+
| name  | age  |
+-----+-----+
| Atomic | 19   |
| DarkLord | 19  |
+-----+-----+
2 rows in set (0.00 sec)
```

9. Write a JDBC program to delete the employee record from Employee table with prepared statement using the following GUI. (HW)

CODE:

```
package jdbc.projects;
import java.sql.*;

public class prepared_employee_delete extends javax.swing.JFrame {
    private static final java.util.logging.Logger logger =
        java.util.logging.Logger.getLogger(prepared_employee_delete.class.getName());
    public prepared_employee_delete() {
        initComponents();
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {
        jLabel1 = new javax.swing.JLabel();
        jButton1 = new javax.swing.JButton();
        jTextField1 = new javax.swing.JTextField();
        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        jLabel1.setBackground(new java.awt.Color(204, 204, 255));
        jLabel1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
        jLabel1.setForeground(new java.awt.Color(102, 102, 255));
        jLabel1.setText("Employee ID:");
        jButton1.setBackground(new java.awt.Color(204, 204, 255));
        jButton1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
        jButton1.setForeground(new java.awt.Color(102, 102, 255));
        jButton1.setText("Submit");
        jButton1.setBorder(javax.swing.BorderFactory.createEtchedBorder());
        jButton1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton1ActionPerformed(evt);
            }
        });

        jTextField1.setBackground(new java.awt.Color(204, 204, 255));
        jTextField1.setFont(new java.awt.Font("Segoe UI", 1, 14)); // NOI18N
        jTextField1.setForeground(new java.awt.Color(102, 102, 255));
        jTextField1.setBorder(javax.swing.BorderFactory.createEtchedBorder());
        javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addGroup(layout.createSequentialGroup()
                    .add(jButton1)
                    .add(jTextField1)
                    .addContainerGap())
        );
    }
}
```

```

        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup())
        .addGap(55, 55, 55)
        .addComponent(jLabel1, javax.swing.GroupLayout.PREFERRED_SIZE, 99, javax.swing.GroupLayout.PREFERRED_SIZE)
        .addGap(45, 45, 45)
        .addComponent(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, 110,
javax.swing.GroupLayout.PREFERRED_SIZE))
        .addGroup(layout.createSequentialGroup())
        .addGap(133, 133, 133)
        .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 99,
javax.swing.GroupLayout.PREFERRED_SIZE)))
        .addContainerGap(119, Short.MAX_VALUE))
    );
    layout.setVerticalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup())
        .addGap(85, 85, 85)
        .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
        .addComponent(jLabel1)
        .addComponent(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
        .addGap(41, 41, 41)
        .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE, 31, javax.swing.GroupLayout.PREFERRED_SIZE)
        .addContainerGap(145, Short.MAX_VALUE))
    );
    pack();
}
private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
    try{
        Class.forName("com.mysql.jdbc.Driver");
        try(Connection con = DriverManager.getConnection("jdbc:mysql://localhost/atomic",
"root", "atomic")){
            String eid = jTextField1.getText();
            String sql1 = "delete from employee where eno='"+ eid +"'";
            PreparedStatement ps = con.prepareStatement(sql1);
            ps.executeUpdate();
            System.out.println("Deleted!!!");
            ps.close();
            con.close();
        }
    }
    catch(ClassNotFoundException | SQLException | NumberFormatException e){
        System.out.println("Error:" + e.getMessage());
    }
}
public static void main(String args[]) {
    java.awt.EventQueue.invokeLater(() -> new prepared_employee_delete().setVisible(true));
}
private javax.swing.JButton jButton1;
private javax.swing.JLabel jLabel1;
private javax.swing.JTextField jTextField1;
}

```

OUTPUT:

Employee ID:

Loading class 'com.mysql.jdbc.Driver'.
Deleted!!

```
mysql> select * from employee;
+-----+-----+-----+-----+
| eno  | salary | department | password |
+-----+-----+-----+-----+
| E001 | 35000  | Sales      | 123      |
| E002 | 45000  | HR         | 456      |
| E003 | 55000  | IT         | 7878     |
| E004 | 60000  | Finance    | 101      |
| E005 | 40000  | Admin      | 540      |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> select * from employee;
+-----+-----+-----+-----+
| eno  | salary | department | password |
+-----+-----+-----+-----+
| E001 | 35000  | Sales      | 123      |
| E003 | 55000  | IT         | 7878     |
| E004 | 60000  | Finance    | 101      |
| E005 | 40000  | Admin      | 540      |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```